



Spinning binaries in scalar-tensor EFTs and their UV consistency from Scattering Amplitudes

Panagiotis Marinellis

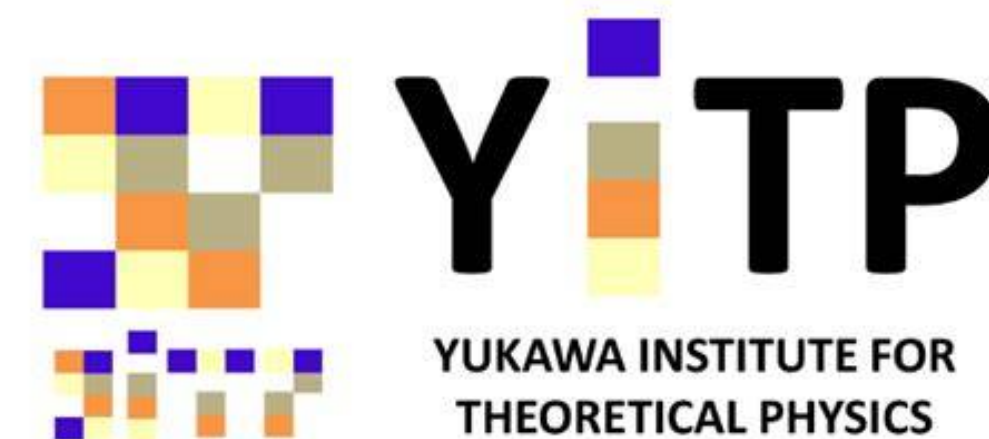
Based on:

[2407.16457] [Falkowski, PM]

[2411.12909] [Falkowski, PM]

[2507.11426] [Alviani, Falkowski, PM]

[26XX.XXXXXX] [PM]



Plan for this talk:

1. Motivation

2. Scalar-tensor EFTs in binaries from Amplitudes

[2407.16457]

Eur.Phys.J.C 85
(2025) 1, 74

[2411.12909]

JHEP 05
(2025) 045

[26XX.XXXXXX]

3. UV completions of scalar-tensor EFTs

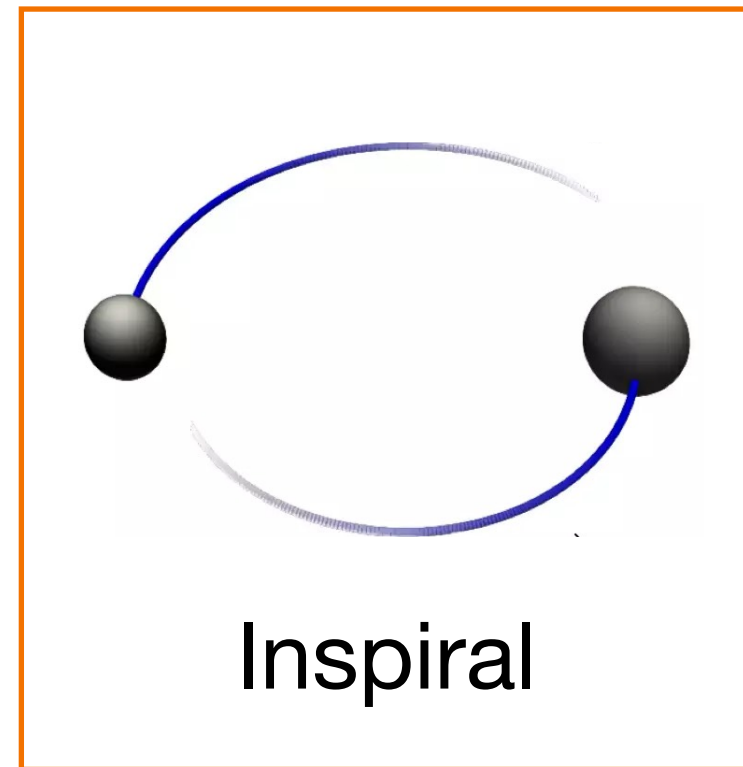
[2507.11426]

JHEP 01
(2026) 060

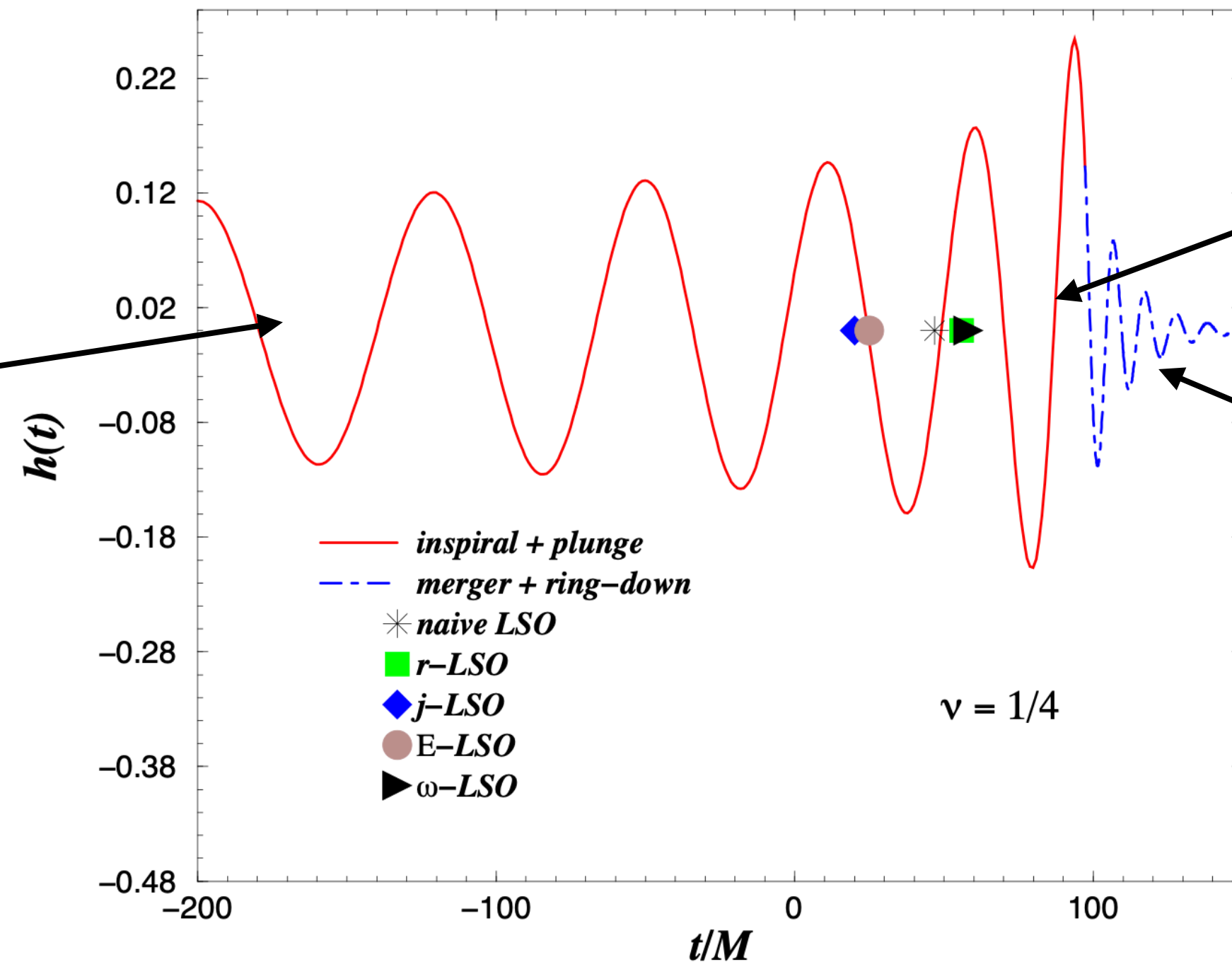
4. Outlook

1. Motivation

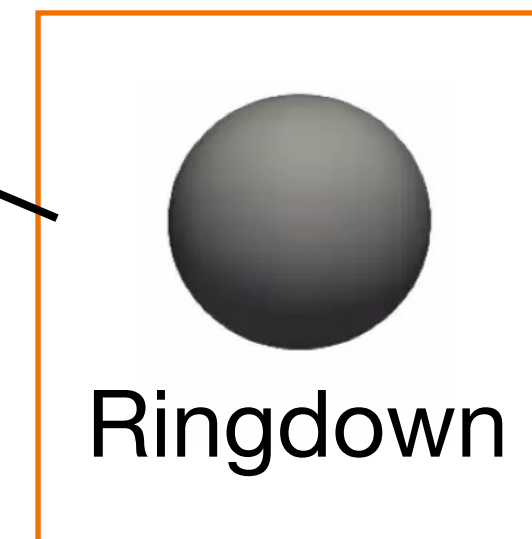
Test effects beyond GR with gravitational waves



Focus specifically on **PM expansion**; Expansion only around weak gravitational fields, $G \ll 1$



Phys.Rev.D 62 (2000) 064015 [Buonanno, Damour]



Scattering information can be used for the **bound** problem!

See talks by Fei, Thibault, Zvi!



What is the nature of these compact objects?

1. Motivation

Black hole hair?

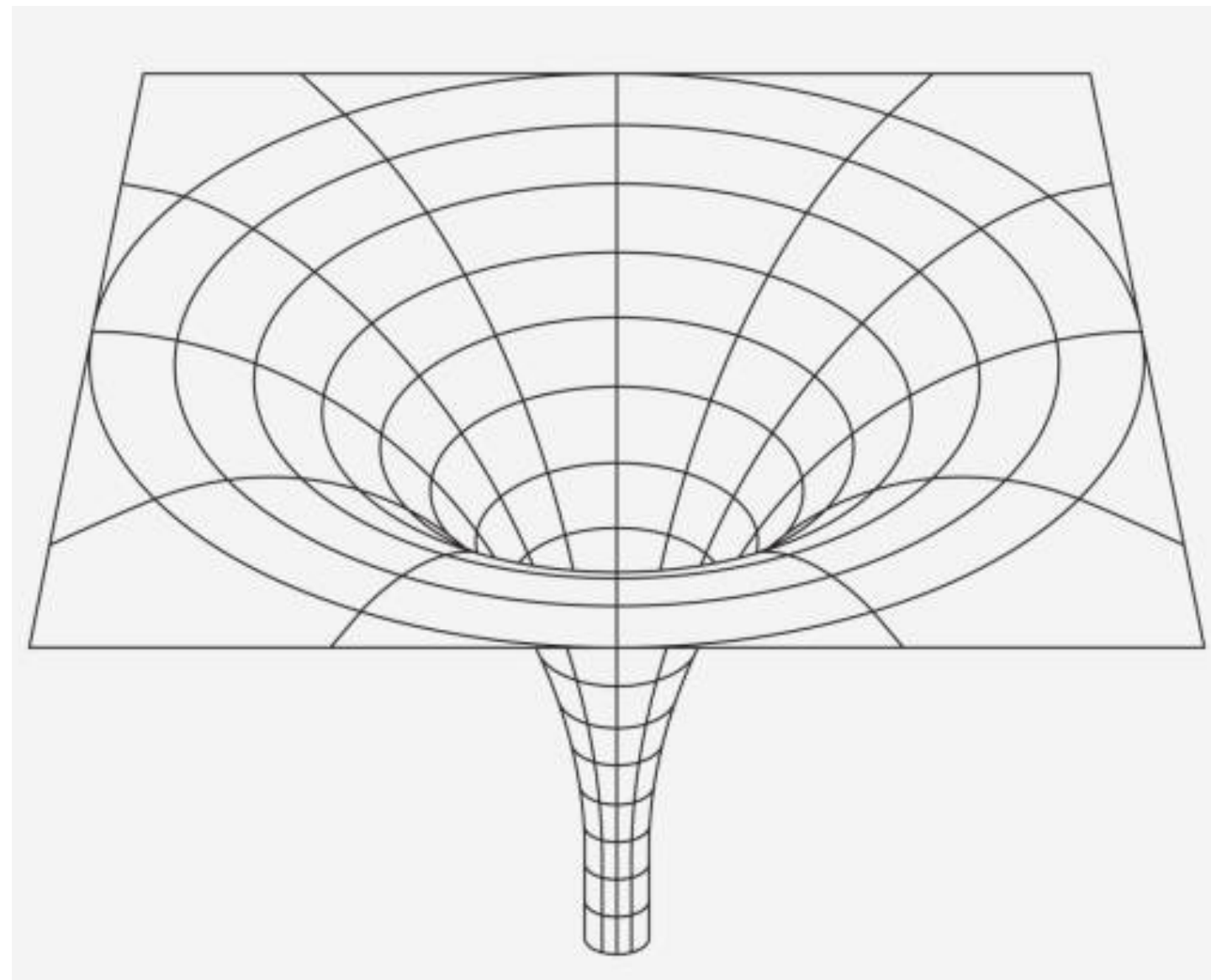
Charge



Mass



Spin



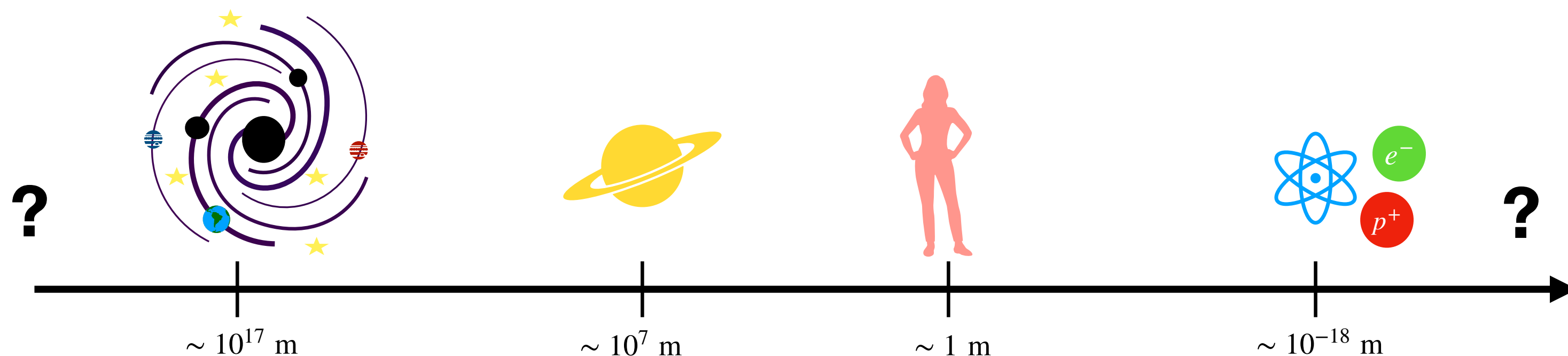
Charge, mass, spin + ?

**Are black holes
characterised by
these quantities
alone?**

2. Scalar-tensor EFTs

- **Scalar-tensor theories** have long stood as a promising avenue to study **extensions of GR**.
- They consist of gravity theories with the introduction of an additional **massless scalar** degree of freedom, along with the **massless graviton**.
- The action of scalar-tensor theories can be systematically studied in the context of effective field theories (**EFTs**); one then builds consistently the **scalar-tensor EFT**.

$$S_{GR}[g_{\mu\nu}] \rightarrow S_{ST}[g_{\mu\nu}, \phi]$$



The core idea of EFTs: at various scales we use different physical theories to describe the relevant phenomena.

2. Scalar-tensor EFTs

$$R^{\mu\nu\rho\sigma}\tilde{R}_{\mu\nu\rho\sigma} \quad , \quad \tilde{R}^{\mu}_{\nu\rho\sigma} = \frac{1}{2}\varepsilon_{\rho\sigma}^{\alpha\beta}R^{\mu}_{\nu\alpha\beta}$$

$$\mathcal{L}_{\text{EFT}} = \sqrt{-g} \left\{ \mathcal{L}_0 + \sum_{D \geq 5} \mathcal{L}^{(D)} \right\}$$

$$\mathcal{L}_0 = \frac{M_{Pl}^2}{2}R + \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{C_3}{3!}\phi^3 - \frac{C_4}{4!}\phi^4 \quad D \leq 4$$

$$R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma} - 4R^{\mu\nu}R_{\mu\nu} + R^2$$

$$\mathcal{L}^{(5)} = \frac{M_{Pl}}{\Lambda^2}\phi \left\{ \alpha \mathcal{R}_{\text{GB}}^2 + \tilde{\alpha} \mathcal{R}_{\text{CS}}^2 \right\} - \frac{C_5}{5!}\phi^5$$

SGB **DCS**

$$\mathcal{L}^{(6)} = \frac{1}{\Lambda^2} \left\{ \frac{C_{R^3}}{3!}\mathcal{R}^3 + \frac{\tilde{C}_{R^3}}{3!}\widetilde{\mathcal{R}}^3 + \phi^2 \left[C_{\phi^2\text{GB}}\mathcal{R}_{\text{GB}}^2 + C_{\phi^2\text{CS}}\mathcal{R}_{\text{CS}}^2 \right] - \frac{C_6}{6!}\phi^6 \right\}$$

...

2. Scalar-tensor EFTs

$$R^{\mu\nu\rho\sigma}\tilde{R}_{\mu\nu\rho\sigma} \quad , \quad \tilde{R}^{\mu}_{\nu\rho\sigma} = \frac{1}{2}\epsilon_{\rho\sigma}^{\alpha\beta}R^{\mu}_{\nu\alpha\beta}$$

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$$\mathcal{L}_{\text{SVT}}^{(5)} \supset \frac{M_{Pl}}{\Lambda^2}\phi \left\{ \beta F^2 + \tilde{\beta} F\tilde{F} \right\}$$

For the extension to EMDA see [26XX.XXXXXX]

Ignoring scalar self interactions ϕ^n

Experimental window:

$$\frac{\sqrt{\alpha}}{\Lambda} \lesssim 0.22km \quad , \quad \frac{\sqrt{\tilde{\alpha}}}{\Lambda} \lesssim 9.5km$$

Broad window of resolvability! Eur.Phys.J.C 85 (2025) 1, 74 [Falkowski, PM]

Phys.Rev.D 107 (2023) 4, 044030 [Silva, Ghosh, Buonanno]
 Phys.Rev.D 111 (2025) 2, 024016 [Julié, Pompili, Buonanno]
 Phys.Rev.Lett. 126 (2021) 18, 181101 [Silva, Holgado, Cárdenas-Avendaño, Yunes]

Already GW observations are used to constrain SGB, DCS

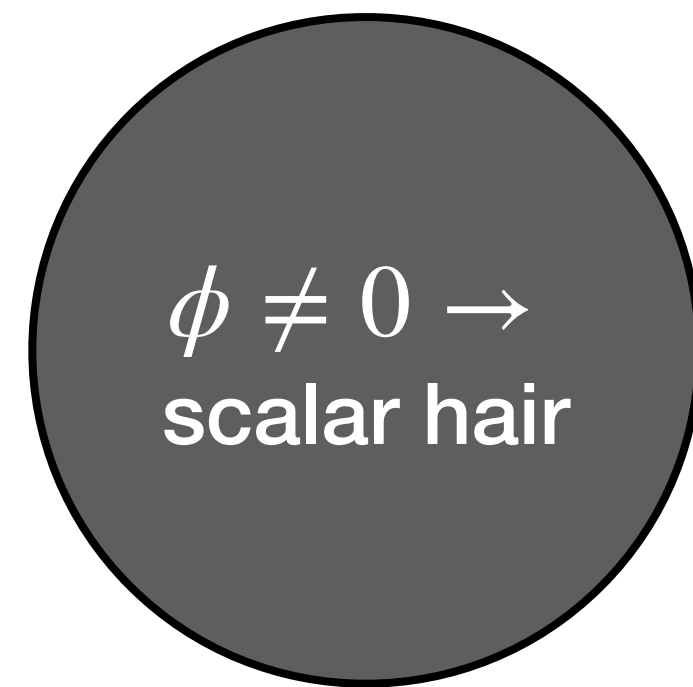
Phys.Rev.D 112 (2025) 4, 044004 [Wilson-Gerow]
 [2601.17503] [Bernard, Jain, Mougiakakos]
 [2602.08876] [Almeida, Du, Liu, Wang]

Recent activity on the PM dynamics using worldline methods!



Scalar hair in scalar-tensor theories:

Compact objects can acquire scalar hair in ST theories $\longrightarrow$ Exactly the case for SGB, DCS



Hairy BH solution in ST theory

Far zone $x \rightarrow \infty$

“Monopole hair” $\swarrow$ $\searrow$ “Dipole hair”

$$\phi = \frac{c_i}{r} + \frac{d_i}{r^2} + \dots$$

SGB $\nearrow$ $\nwarrow$ **DCS**

Most **phenomenologically relevant** effect

Monopole leads to **dipole radiation** for binary inspirals

$$\frac{P_{\phi}^{\text{LO}}}{P_{GR,h}^{\text{LO}}} \sim \frac{1}{v^2}$$

Phys.Rev.D 85 (2012) 064022,
Phys.Rev.D 93 (2016) 2, 029902
(erratum) [Yagi, Stein, Yunes, Tanaka]

Shift-symmetry of the SGB theory plays a crucial role to **avoid no-hair theorems** for scalar-tensor theories

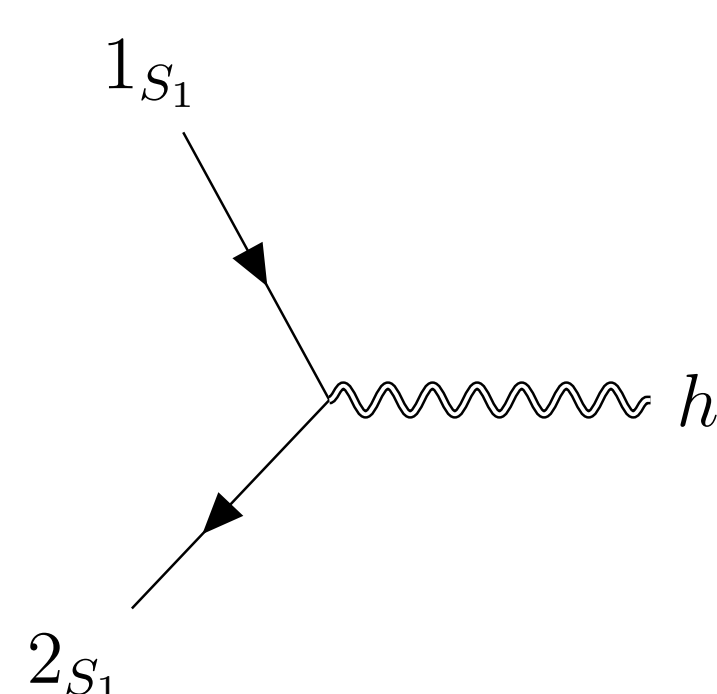
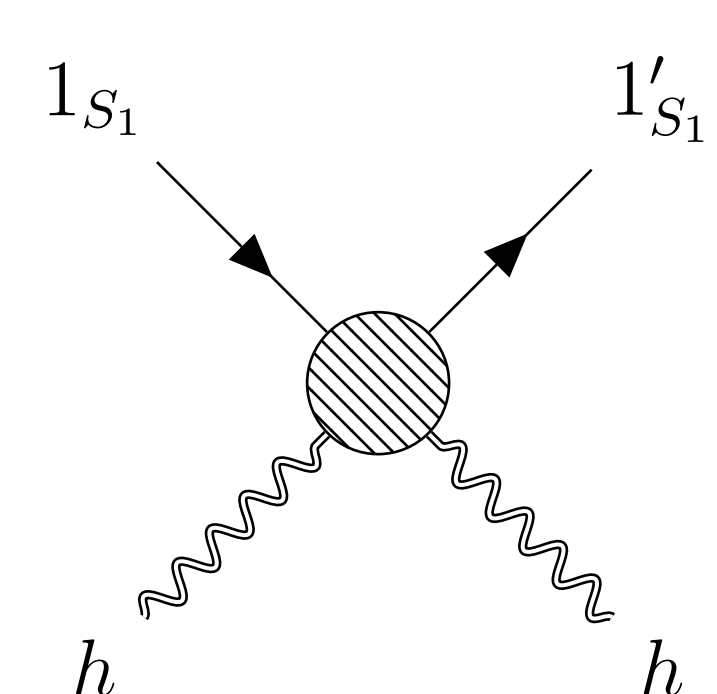
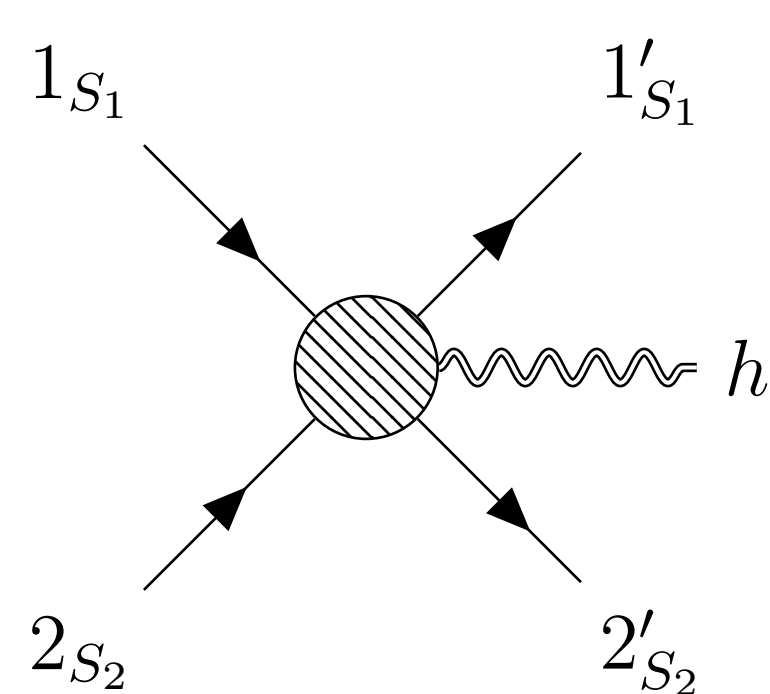
Phys.Rev.D 99 (2019) 12, 124004
[Saravani, Sotiriou]

Effects of higher order terms can be important,
e.g. $\phi^2 \mathcal{R}_{GB}^2$ leads to **spontaneous scalarization**

Phys.Rev.Lett. 120 (2018) 13,
131103 [Doneva, Yazadjiev]
Phys.Rev.Lett. 120 (2018) 13, 131104
[Silva, Sakstein, Gualtieri, Sotiriou, Berti]



Waveforms at leading order with KMOC:

3-point building blocks	Modifications to 4-points + Contact terms' ansatz! See Lucile's talk!	5-points (and waveforms)
		

strain

$$h_{GR}(t)$$

$$\equiv h_+ \pm i h_\times$$

$$\sim$$

$$\int$$

$$d\omega$$

$$e^{-i\omega t}$$

$$\int$$

$$d\Phi(q)$$

$$e^{-ibq}$$

on-shell measure

$$1_{S_1}$$

$$2_{S_2}$$

$$1'_{S_1}$$

$$2'_{S_2}$$

$$h$$

$$\mathcal{M}_5^{\text{cl}}(q, k) \big|_{k^\mu = \omega n^\mu}$$

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See talks by Donal and Fei!

Inclusion of scalar-tensor interactions

Universal interactions with the $\mathcal{R}_{\text{GB}}^2$ and $\mathcal{R}_{\text{CS}}^2$ invariants come from the dimension-5 operators

$$\mathcal{L}^{(5)} = \frac{M_{Pl}}{\Lambda^2} \phi \left\{ \alpha \mathcal{R}_{\text{GB}}^2 + \tilde{\alpha} \mathcal{R}_{\text{CS}}^2 \right\} \xrightarrow{g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}} \begin{array}{c} h \\ \text{wavy line} \\ h \end{array} \cdots \phi \sim \frac{\hat{\alpha}}{\Lambda^2} = \frac{\alpha + i\tilde{\alpha}}{\Lambda^2}$$

All compact objects will radiate the scalar (and GWs) through this extra coupling



At leading **PN** order

$$\left. \frac{dP_\phi}{d\Omega} \right|_{\mathcal{O}(a^0)} \sim \frac{v^6}{b^8} \xrightarrow{\text{Kepler's law}} \left. \frac{dP_\phi}{d\Omega} \right|_{\mathcal{O}(a^0)} \sim v^{22}$$

Heavily suppressed compared to “hairy” compact objects

On-shell description of monopole hair

Phys.Rev.D 48 (1993)
3641-3647 [Bekenstein]

We model the BH as a point-particle interacting with the scalar field in a ST fashion

Neglect it,
suppressed

Most general effective metric that respects causality is:

$$\tilde{g}_{\mu\nu} = \underbrace{\exp\left[C\left(\frac{\phi}{M_{Pl}}\right)\right]}_{\text{Conformal coupling}} g_{\mu\nu} + \underbrace{D\left(\frac{\phi}{M_{Pl}}\right)}_{\text{Disformal coupling}} \underbrace{\frac{D_\mu\phi D_\nu\phi}{M_{Pl}^2\Lambda^2}}_{\text{Disformal coupling}},$$

Generate 3-point amplitudes for arbitrary spinning BH:

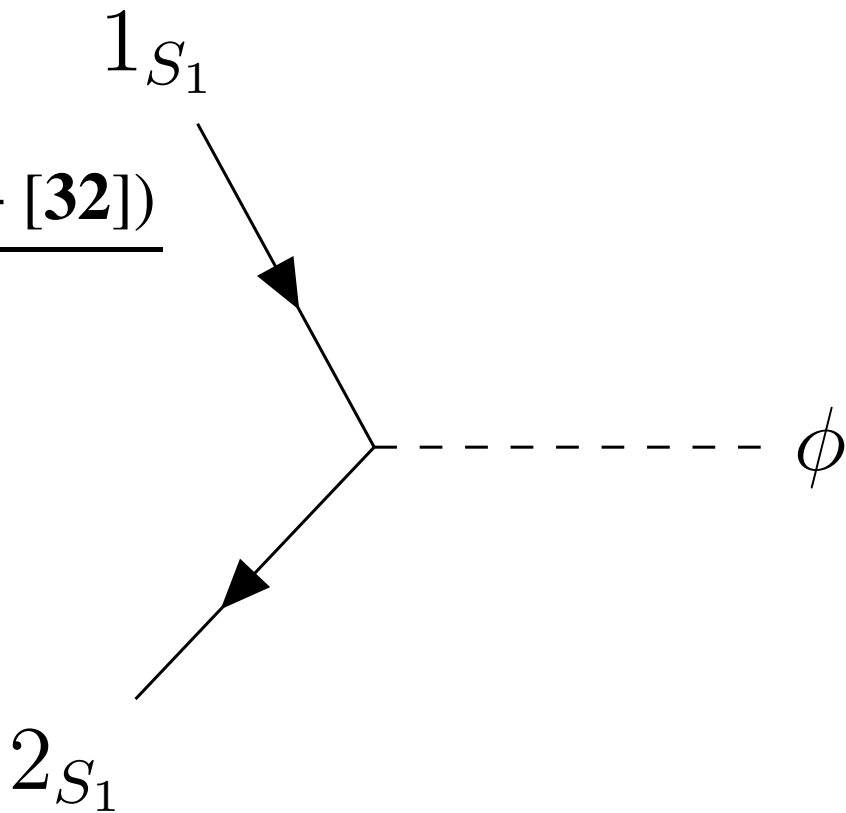
$$\exp\left[C\left(\frac{\phi}{M_{Pl}}\right)\right] \approx 1 + c\frac{\phi}{M_{Pl}}$$

$$\mathcal{M}_{3,bos.}(1_\phi, 2_A, 3_A) = \frac{c_1}{M_{Pl}} \frac{\langle 32 \rangle^S [32]^S}{m^{2S-2}}$$

$$\mathcal{M}_{3,ferm.}(1_\phi, 2_\Psi, 3_{\bar\Psi}) = \frac{c_1}{M_{Pl}} \frac{\langle 32 \rangle^{S-1/2} [32]^{S-1/2} (\langle 32 \rangle + [32])}{m^{2S-2}}$$

Simple mass redefinition
at all spin orders:

$$m \rightarrow e^{C/2} m$$



Resemblance to skeletonized action know long ago:

Astrophysical Journal, vol. 196, Mar. 1, 1975, pt. 2, p. L59-L62. [Eardley]
1992 Class. Quantum Grav. 9 2093 [Damour, Esposito-Farese]

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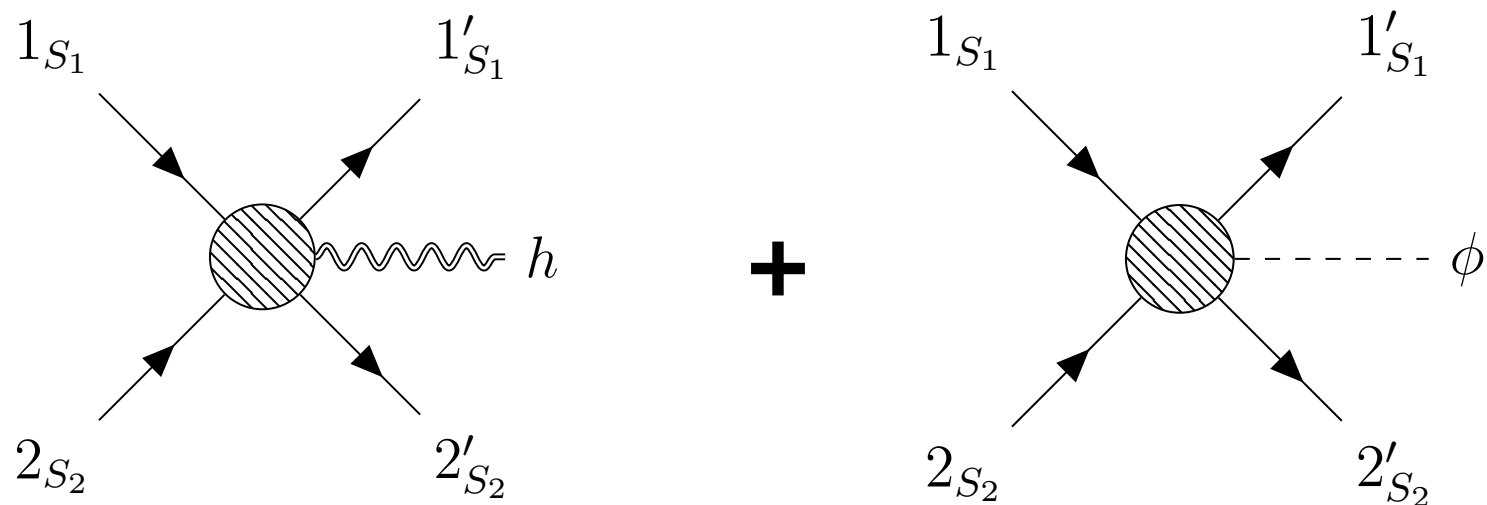
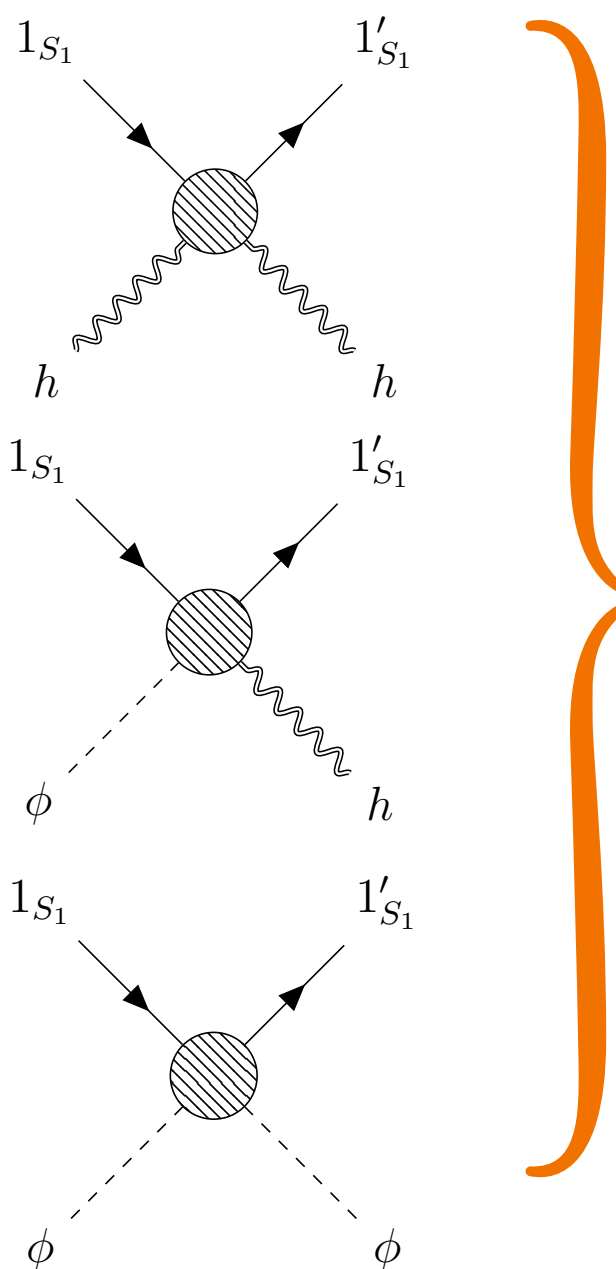
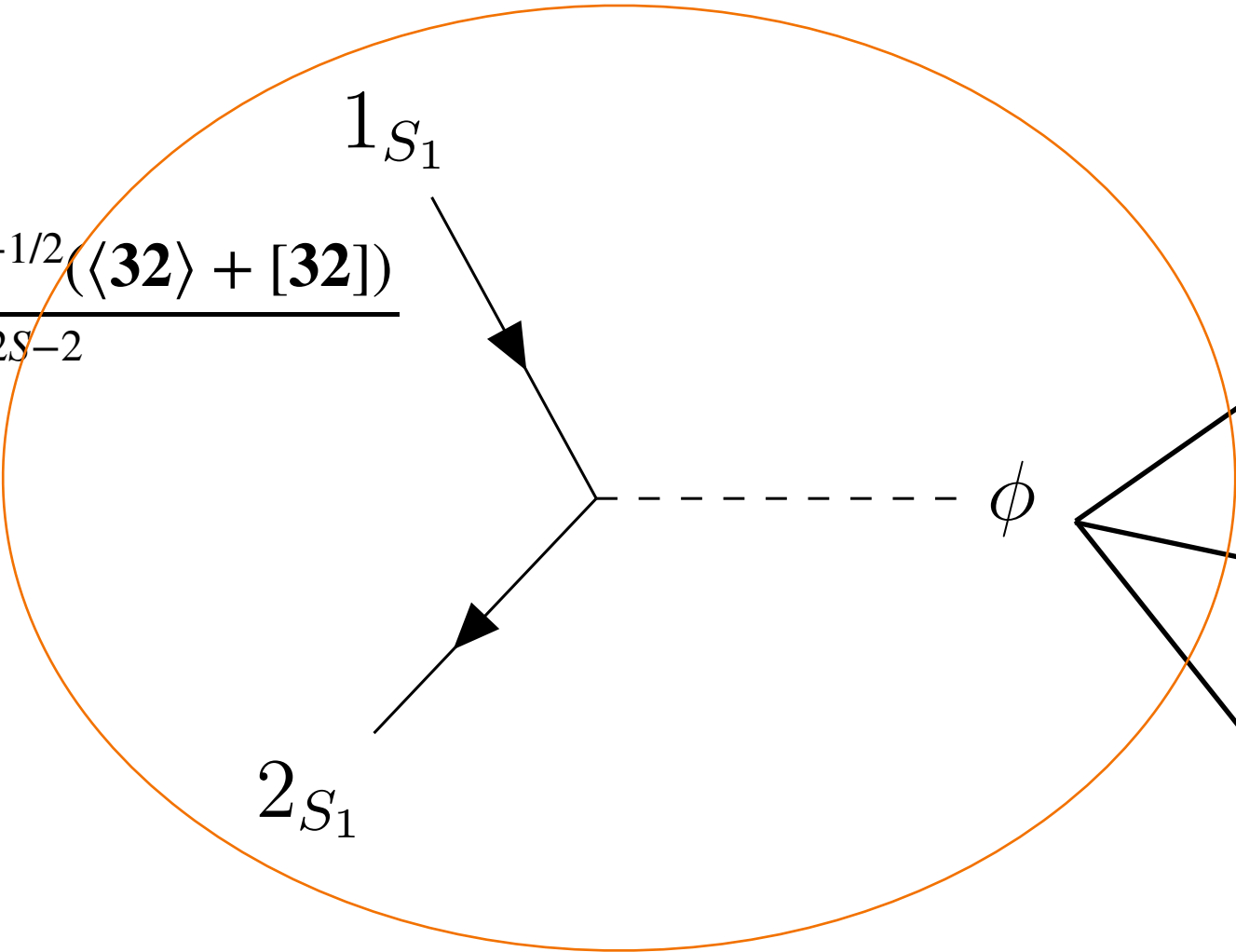
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1992 Class. Quantum Grav. 9 2093 [Damour, Esposito-Farese]

Waveforms and emitted power

Waveforms for scattering of compact objects with scalar-hair:

$$c_i \sim \alpha + \mathcal{O}(a^2)$$

Phys.Rev.D 100 (2019) 10, 104061 [Julié, Berti]

LO Scalar Waveforms-Spinless part:

scalar “monopole” charges $c_i = c_i(m_i, a_i, \frac{\alpha}{\Lambda^2}) \longrightarrow$ Can be matched to existing results if it’s secondary hair!

$$W_\phi^{(0)} = -\frac{m_1 m_2}{32 \pi^2 M_{Pl}^3 \sqrt{\gamma^2 - 1} (\hat{u}_1 n)^2 b} \frac{1}{\sqrt{1 + T_1^2}} \left\{ \frac{(\gamma^2 - 1) [c_1 (\hat{u}_2 n)^2 + c_2 (\hat{u}_1 n)^2] [\gamma (\hat{u}_2 n) - (\hat{u}_1 n) + (\tilde{b} n) T_1]}{[-(\hat{u}_1 n) + \gamma (\hat{u}_2 n) + (\tilde{b} n) T_1]^2 + (\tilde{v} n)^2 (1 + T_1^2)} - \frac{c_1 (\hat{u}_1 n) + (2\gamma^2 - 3) y (\hat{u}_2 n) - (2\gamma^2 - 1) (\tilde{b} n) T_1}{\gamma^2 - 1} + \frac{C_1^{(0)}}{2} c_2 (\hat{u}_1 n) \right\} + (1 \leftrightarrow 2).$$

Zeroth spin order contact term

$v \ll 1$ limit

$$\left. \frac{dP_\phi}{d\Omega} \right|_{\mathcal{O}(a^0)} \sim \frac{1}{b^4}$$

Same for the gravitational waveforms:

$$\left. \frac{dP_h}{d\Omega} \right|_{\mathcal{O}(a^0)} \sim \frac{v^2}{b^4}$$

Kepler’s law

$$\left. \frac{dP_\phi}{d\Omega} \right|_{\mathcal{O}(a^0)} \sim v^8$$

Kepler’s law

$$\left. \frac{dP_h}{d\Omega} \right|_{\mathcal{O}(a^0)} \sim v^{10}$$

Agreement with existing PN results for SGB!

Connect to observables:
Power emitted in scalar radiation

$$\frac{dP_\phi}{d\Omega} = (\partial_t W_\phi)^2$$

Phys.Rev.D 85 (2012) 064022, Phys.Rev.D 93 (2016) 2, 029902 (erratum) [Yagi, Stein, Yunes, Tanaka]
Class.Quant.Grav. 39 (2022) 3, 035002 [Shiralilou, Hinderer, Nissanke, Ortiz, Witek]

JHEP 05 (2025) 045 [Falkowski, PM]

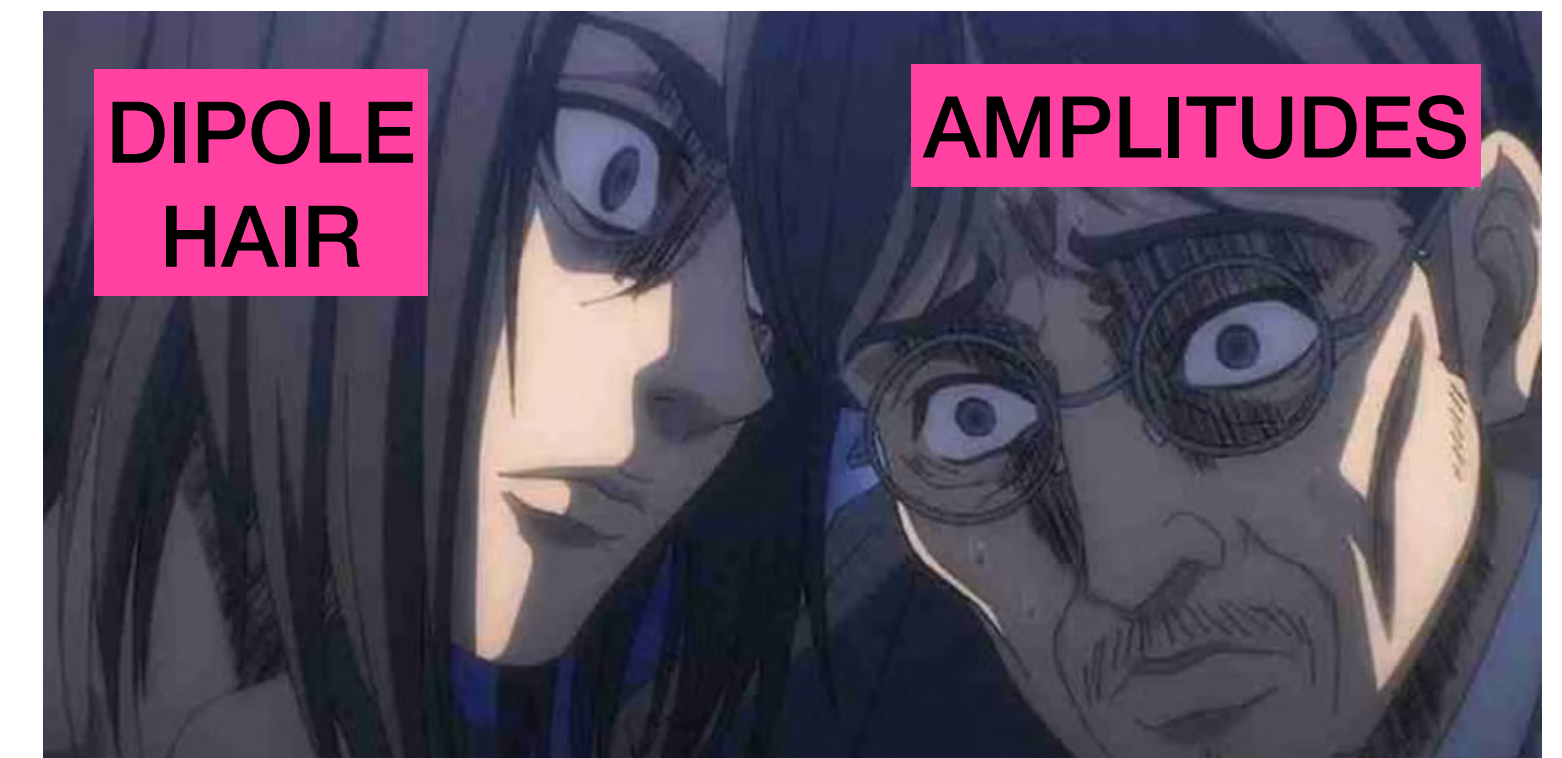
On-shell description of dipole hair?

However, even after parametrizing our ignorance of the amplitudes we were still not able to reproduce the on-shell description of the (spin-dependent) dipole hair!

We can only reproduce part of the result and only in a particular limit. We get:

$$P_{\phi}^{(0)} \rightarrow 0$$

$$P_{\phi}^{(2)} \rightarrow \frac{m_1^2 m_2^2}{1024 \pi^3 M_{Pl}^6 b^6} \left([c_1 C_2^{(1)} \mathbf{a}_2 - c_2 C_1^{(1)} \mathbf{a}_1] \cdot \hat{\mathbf{v}} \right)^2 v^2$$



The leading PN formula for **DCS** reads:

$$P_{\phi}^{(2)} \sim (\mathbf{a} \cdot \mathbf{v})^2 + \# \mathbf{a}^2 \mathbf{v}^2$$

Phys.Rev.D 85 (2012) 064022, Phys.Rev.D 93 (2016) 2, 029902 (erratum) [Yagi, Stein, Yunes, Tanaka]

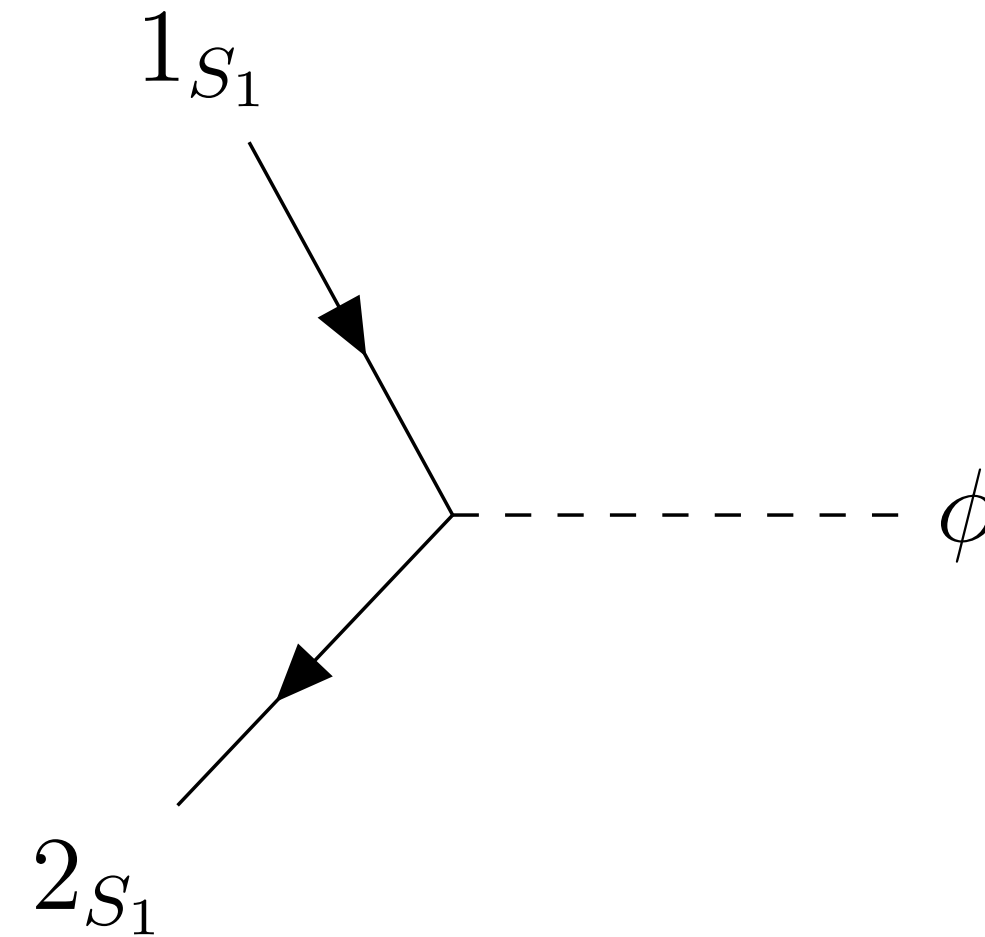
We couldn't reproduce the second term...!

Any **new ideas**? Please come **talk to me!!!**
(All communities welcome!)

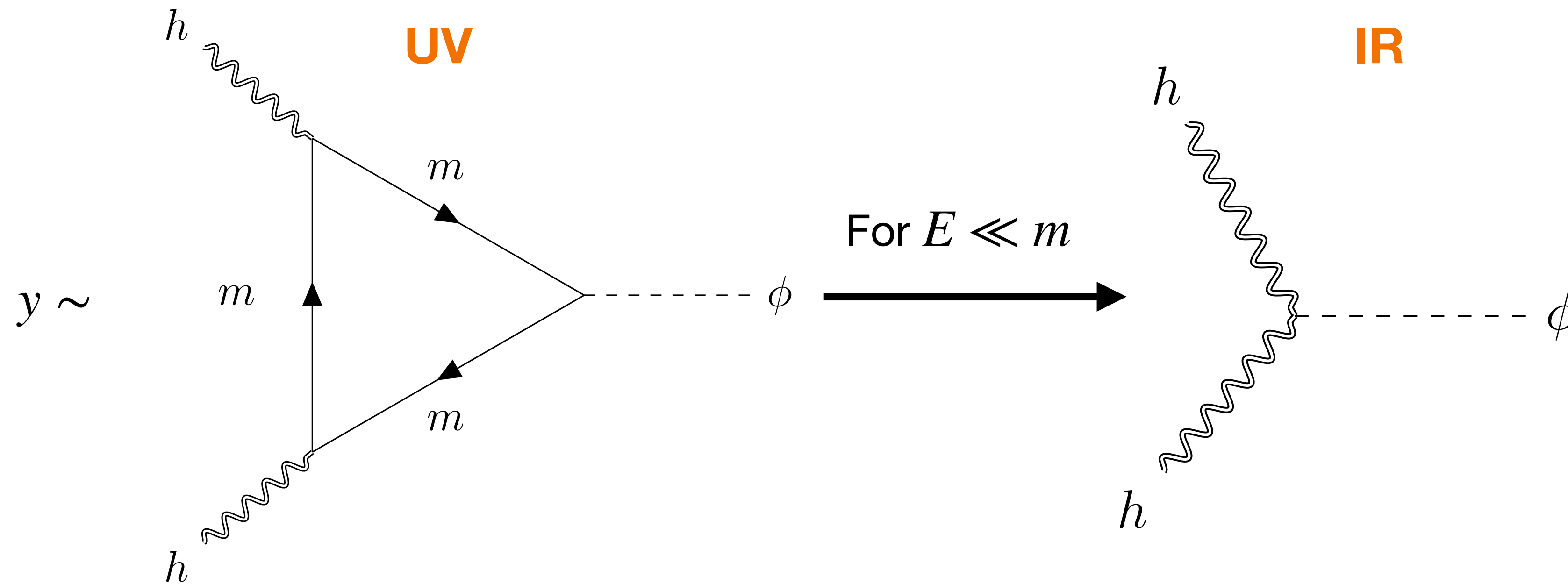
4. UV completions of scalar-tensor EFTs

JHEP 01 (2026) 060
[Falkowski, PM]

Idea:



$\sim y$
coupling of **massive matter**
to **massless scalar**



$\sim C_{\phi\text{GB}}(y, m) \pm iC_{\phi\text{CS}}(y, m)$
SGB/DCS interaction



4. UV completions of scalar-tensor EFTs

Parametrisation of all possible interactions with massive particle with spin $S = 0, 1/2, 1$

$$\mathcal{L}_{\text{UV}}^{\text{cubic}} \supset -\phi \left\{ y_0 m_\Phi |\Phi|^2 + y_s (\bar{\Psi}\Psi) - i y_p (\bar{\Psi}\gamma^5\Psi) + \frac{1}{M_*} \left[y_d V_{\mu\nu} \bar{V}^{\mu\nu} + y_a \widetilde{V}_{\mu\nu} \bar{V}^{\mu\nu} - y_m m^2 V_\mu \bar{V}^\mu \right] \right\}$$

3-points

$$\mathcal{M}[1_\phi 2_\Phi 3_{\bar{\Phi}}] = -y_0 m_\Phi \quad \mathcal{M}[1_\phi 2_\Psi 3_{\bar{\Psi}}] = -y_s [\langle \mathbf{23} \rangle + [\mathbf{23}]] - i y_p [\langle \mathbf{23} \rangle - [\mathbf{23}]]$$

$$\mathcal{M}[1_\phi 2_V 3_{\bar{V}}] = -\frac{1}{M_*} \left\{ y_d [\langle \mathbf{23} \rangle^2 + [\mathbf{23}]^2] + i y_a [\langle \mathbf{23} \rangle^2 - [\mathbf{23}]^2] + y_m \langle \mathbf{23} \rangle [\mathbf{23}] \right\}$$

Inclusion of leading EFT quadratic operators, aka contact terms

$$\mathcal{L}_{\text{UV}}^{\text{quadratic}} \supset -\frac{\phi^2}{2} \left\{ \frac{c_\Phi}{2} m_\Phi |\Phi|^2 - \frac{c_\Psi}{M_*} (\bar{\Psi}\Psi) + i \frac{\tilde{c}_\Psi}{M_*} (\bar{\Psi}\gamma^5\Psi) - \frac{1}{M_*^2} \left[c_V V_{\mu\nu} \bar{V}^{\mu\nu} + \tilde{c}_V \widetilde{V}_{\mu\nu} \bar{V}^{\mu\nu} - \hat{c}_V m^2 V_\mu \bar{V}^\mu \right] \right\}.$$

On-shell matching

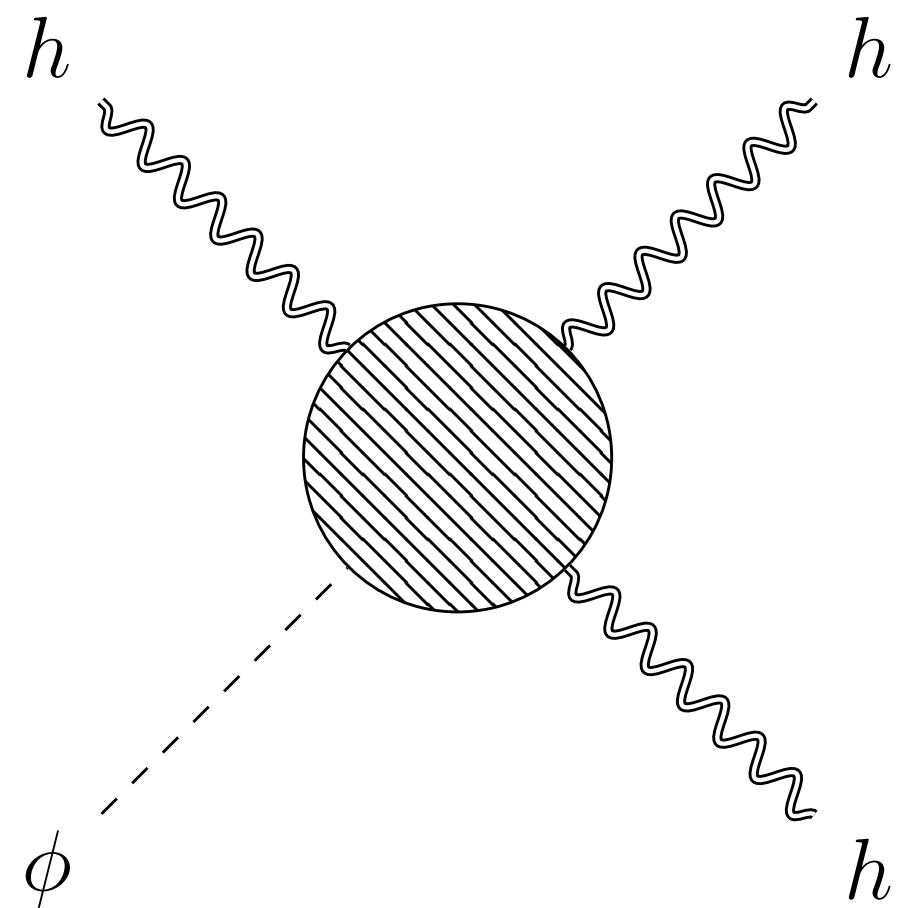


General approach:

Compute **1-loop discontinuities**, e.g.

$$\text{Disc}^s \mathcal{M}^{(1)}[1_h^- 2_h^- 3_h^s 4_\phi] = i \int d\Pi_{XY} \mathcal{M}(1_h^- 2_h^- (-Y)_\Phi (-X)_{\bar{\Phi}}) \mathcal{M}(3_h^s 4_\phi X_\Phi Y_{\bar{\Phi}})$$

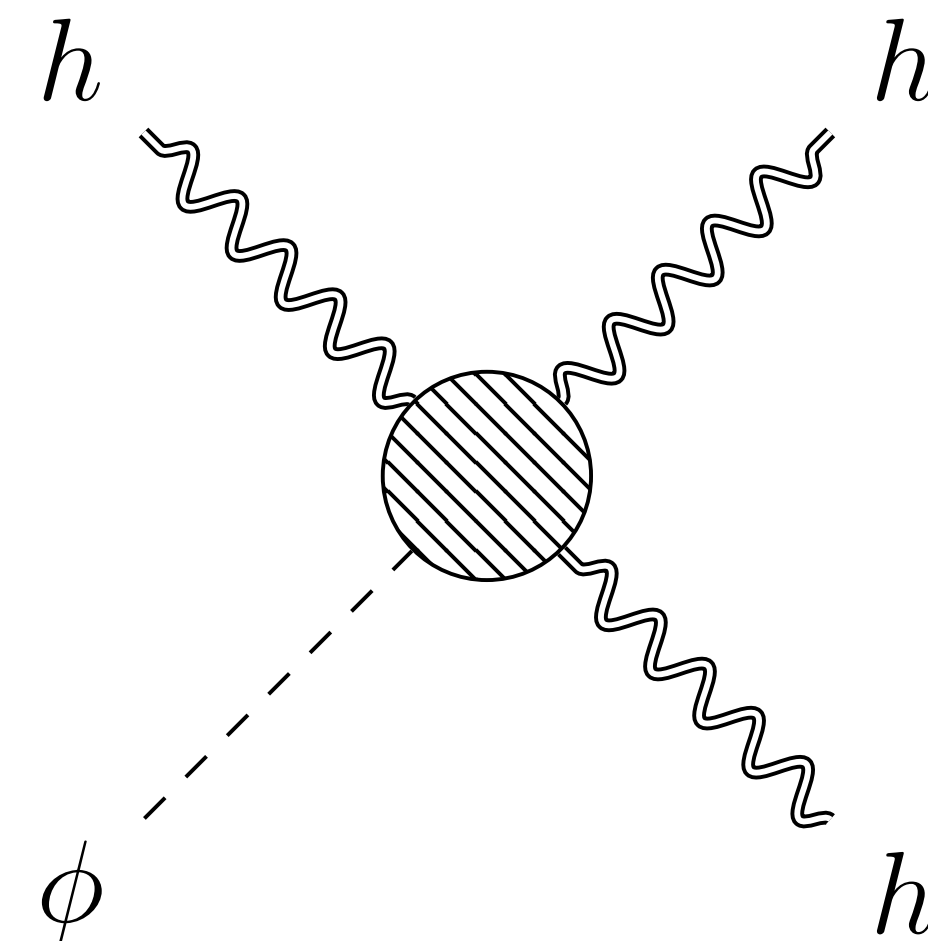
**matter running
inside the loops**



Heavy mass expansion, $m \gg E$



**tree-level EFT
amplitudes**



Some results

JHEP 01 (2026) 060
[Falkowski, PM]



Shift-symmetric SGB/DCS
interactions

$$C_{\phi\text{GB}} = -\frac{y_0}{1440\pi^2m_\Phi} - \frac{7y_s}{2880\pi^2m_\Psi} - \frac{11y_d - 2y_m}{240\pi^2M_*} + \mathcal{O}(M_*^{-1})$$

$$C_{\phi\text{CS}} = -\frac{y_p}{192\pi^2m_\Psi} - \frac{y_a}{24\pi^2M_*} + \mathcal{O}(M_*^{-1})$$

$$C_{\phi^2\text{GB}} = \frac{y_0^2 - c_\Phi}{1440\pi^2m_\Phi^2} + \frac{7(y_s^2 - y_p^2)}{2880\pi^2m_\Psi^2} + \frac{7c_\Psi}{2880\pi^2M_*m_\Psi} + \frac{1}{M_*^2\pi^2} \left\{ \frac{5y_d^2}{96} - \frac{y_m^2}{768} - \frac{19y_a^2}{480} + \frac{11c_V}{240} - \frac{11\hat{c}_V}{1920} \right\} + \mathcal{O}(M_*^{-2})$$

$$C_{\phi^2\text{CS}} = \frac{y_sy_p}{96\pi^2m_\Psi^2} + \frac{\tilde{c}_\Psi}{192\pi^2M_*m_\Psi} + \frac{2y_dy_a + \tilde{c}_V}{24M_*^2\pi^2} + \mathcal{O}(M_*^{-2})$$

$$C_{\phi R^3} = \frac{y_0}{40320\pi^2m_\Phi^3} - \frac{y_s}{10080\pi^2m_\Psi^3} + \frac{2y_d + y_m}{13440\pi^2M_*m_V^2} + \mathcal{O}(M_*^{-3})$$

$$\widetilde{C}_{\phi R^3} = \mathcal{O}(M_*^{-2})$$

$$C_{R^3} = -\frac{1}{40320\pi^2m_\Phi^2} + \frac{1}{20160\pi^2m_\Psi^2} - \frac{1}{13440\pi^2m_V^2} + \mathcal{O}(M_*^{-2})$$

$$\widetilde{C}_{R^3} = \mathcal{O}(M_*^{-2})$$

Wilson coeffs. for R^3 interaction
Crosschecked with

J.Phys.A 54 (2021) 34, 344002 [Bern, Kosmopoulos, Zhiboedov]
Eur.Phys.J.C 85 (2025) 2, 154 [Alviani, Falkowski]

Shift-symmetry
breaking
interactions

Interpretation of the results



JHEP 01 (2026) 060
[Falkowski, PM]

1. UV completions with with spin $S \geq 1$ seem to be **non-informative**
Contributions to Wilson coeffs. suppressed by m_V/M^*
Physics at the scale M^* come at the same order

2. SGB generated, but with **shift-symmetry breaking** operators
May be cancelled but only by fine tuning?

3. DCS can be generated, with a **unique shift-symmetric** choice, corresponding to a theory of **spin 1/2 fermions** and a **complex scalar** with a **Peccei-Quinn global symmetry** $U(1)_{PQ}$.

Lagrangian term

$$\mathcal{L} \supset -\sqrt{2}y_p[H\bar{\Psi}_R\Psi_L + \text{h.c.}]$$

Symmetry
breaking

$$\langle H \rangle = f$$

$$H = \frac{f+h}{\sqrt{2}}e^{i\phi/f}$$

$$H \rightarrow e^{i\alpha}H, \quad \Psi_L \rightarrow e^{-i\alpha}\Psi_L, \quad \Psi_R \rightarrow \Psi_R$$

Action of $U(1)_{PQ}$

$$\mathcal{L} \supset -y_p f \left[e^{-i\phi/f} \bar{\Psi}_R \Psi_L + e^{i\phi/f} \bar{\Psi}_L \Psi_R \right] = -m_\Psi (\bar{\Psi} \Psi) + i y_p \phi (\bar{\Psi} \gamma_5 \Psi) + \underbrace{\frac{y_p^2}{m_\Psi} \frac{\phi^2}{2} (\bar{\Psi} \Psi)}_{\text{Fixing } c_\Psi = \frac{y_p^2 M^*}{m_\Psi}} + \mathcal{O}(\phi^3),$$

$$\text{Fixing } c_\Psi = \frac{y_p^2 M^*}{m_\Psi}$$

$$m_\Psi = y_p f$$

5. Outlook



Conclusions:

- **Amplitudes** methods can efficiently compute (spinning) **classical observables** and perform **on-shell EFT matching**
- The **on-shell** language significantly **simplifies** the calculations, especially when **spin** is involved.

Future opens many questions:

- Employing similar methods to **study GR (and beyond) effects**, where current PN/NR results are poor, e.g. Description of **dipole hair in DCS from amplitudes, dynamical scalarization from amplitudes (?)**
- What is the general **guiding principle** to generate **shift-symmetric** interactions?
- What is the fate of black hole solutions in the presence of the higher dimensional operators generated?
 - Role of $C_{\phi^2\text{GB}}$ for **spontaneous scalarization**
 - Is **scalar hair** still present?
- Beyond our work on UV completions:
 - Do interactions with GB **exponentiate**?
 - Relation with **string theory amplitudes** and UV completions?
 - Does the space of allowed EFT coefficients lie in “**islands**”?
- Could one find a more efficient **on-shell EFT matching** procedure **avoiding 4-points** and only using **3-point information**? Please talk to me for any new ideas!

J.Phys.A 54 (2021) 34, 344002
[Bern, Kosmopoulos, Zhiboedov]

A transformative era in astronomy with GWs!

We can learn a lot from each other!



GR



Thank you for your attention!:)



Backup slides

Causality constraints for SGB and DCS

1. Causality implies that $|\alpha + i\tilde{\alpha}| \lesssim 1$, or $\Lambda \lesssim 1\text{km}^{-1}$ for solar mass BHs, assuming resolvability in the sense of $\frac{|\alpha + i\tilde{\alpha}|}{\Lambda} \sim R_s$.

JHEP 08 (2022) 157[Serra, Serra, Trincherini, Trombetta]

→ Raises question for using these EFTs for the merger dynamics, but still a *broad window where causality is preserved for the inspiral*.

JCAP 08 (2025) 086 [Nie, Tan, Zhang, Zhou]

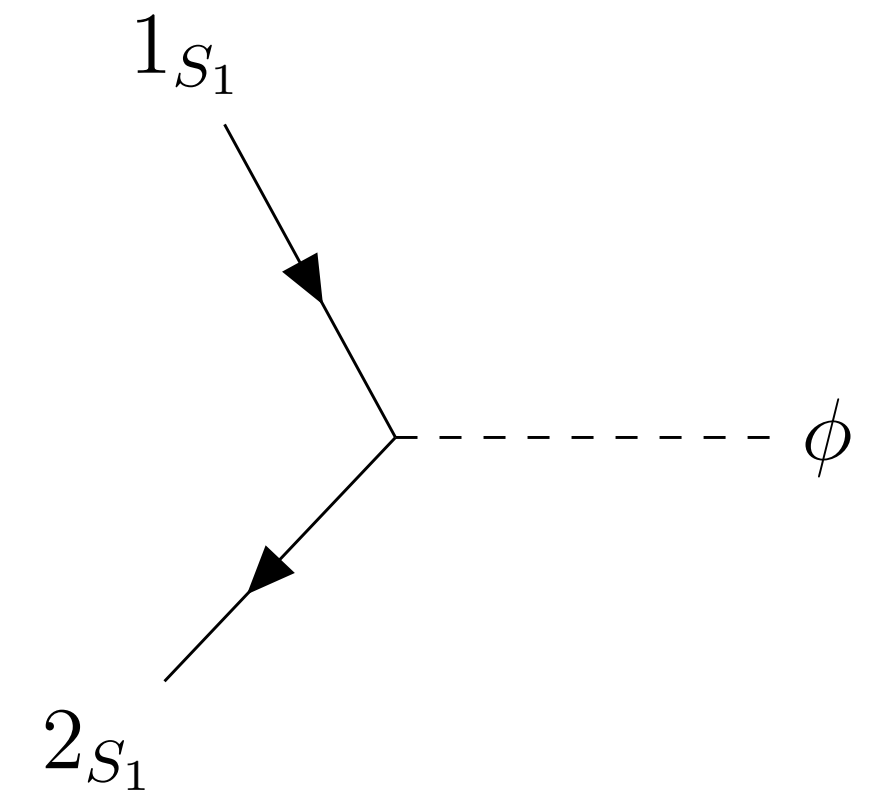
See also:

JHEP 10 (2023) 135 [Hong, Wang, Zhou]

JCAP 01 (2025) 102 [Xu, Hong, Wang, Zhou]

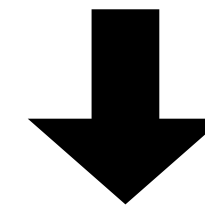
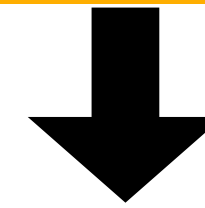
2. However, our “hairy coupling” never considered before: New bounds using this coupling? Is this necessary to draw conclusions for hairy compact objects? Still an open question...

3. In any case, our discussion for monopole scalar charges is *model-independent*, it can be thus used as a useful way to extract templates for (any) spinning compact object that admits scalar hair.



On-shell techniques:

Basic QFT Principles: Poincaré invariance, locality, unitarity of the Scattering Matrix



Scattering Amplitude:
 $\mathcal{M}^{h_1, \dots, h_m}(p_1, \dots, p_n)$

On-shell techniques

Use spinors as variables instead of momenta

Massless spinors: $p_\mu \sigma^\mu_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$
Massive spinors: $p_\mu \sigma^\mu_{\alpha\dot{\alpha}} = \varepsilon_{IJ} \chi^I_\alpha \tilde{\chi}^J_{\dot{\alpha}}$ } Obey little group transformation rules

$$\langle ij \rangle = \varepsilon_{\alpha\beta} \lambda_i^\alpha \lambda_j^\beta, \quad [ij] = \varepsilon_{\dot{\alpha}\dot{\beta}} \tilde{\lambda}_i^{\dot{\alpha}} \tilde{\lambda}_j^{\dot{\beta}}$$

$$\langle \mathbf{ij} \rangle = \langle i^I j^J \rangle, \quad [\mathbf{ij}] = [i^I j^J]$$

On-shell techniques:

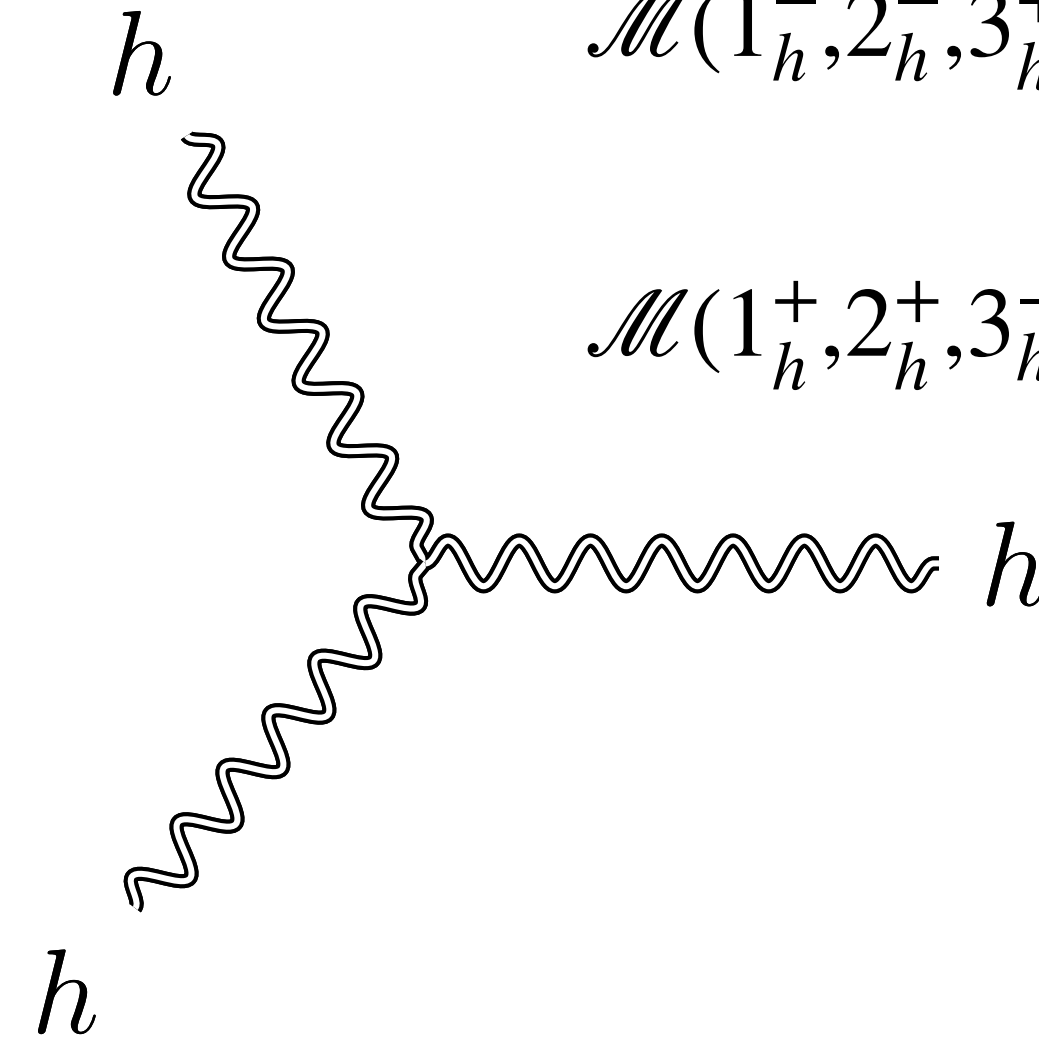
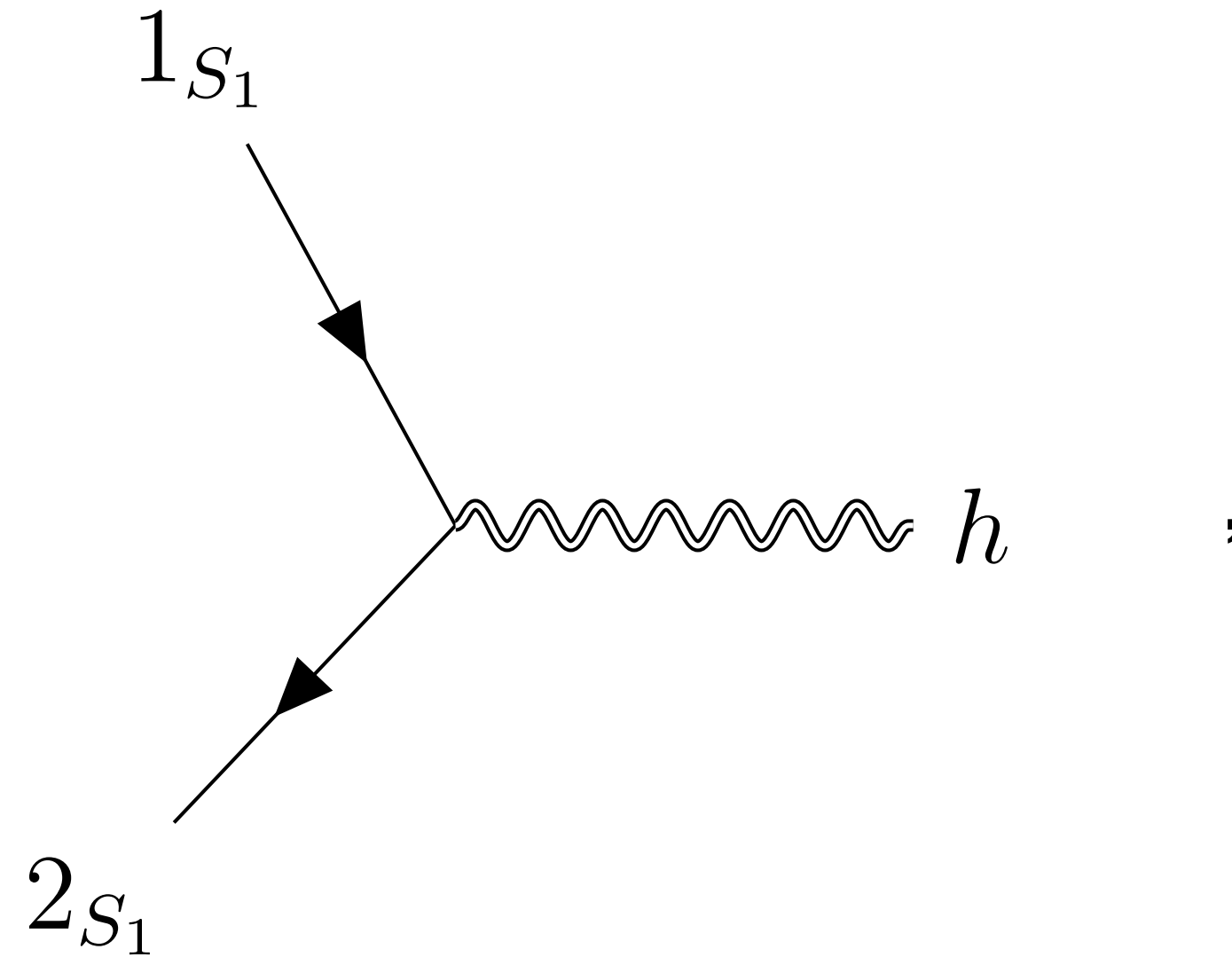


Idea: Build the **on-shell 3-point amplitudes** of the theory

e.g.: **Spinning matter in GR**

$$\mathcal{M}[1_\Phi 2_{\bar{\Phi}} 3_h^-] = -\frac{\langle 3 | p_1 | \tilde{\zeta} \rangle^2 [2\mathbf{1}]^{2S}}{M_{Pl} [3\tilde{\zeta}]^2 m^{2S}},$$

$$\mathcal{M}[1_\Phi 2_{\bar{\Phi}} 3_h^+] = -\frac{\langle \zeta | p_1 | 3 \rangle^2 \langle 2\mathbf{1} \rangle^{2S}}{M_{Pl} \langle 3\zeta \rangle^2 m^{2S}},$$



$$\mathcal{M}(1_h^-, 2_h^-, 3_h^+) = -\frac{1}{M_{Pl}} \frac{\langle 12 \rangle^6}{\langle 13 \rangle^2 \langle 23 \rangle^2}$$

$$\mathcal{M}(1_h^+, 2_h^+, 3_h^-) = -\frac{1}{M_{Pl}} \frac{[12]^6}{[13]^2 [23]^2}.$$

→ Build higher-point amplitudes from their **residues** at kinematic poles in the **complex** plane (up to contact terms)

e.g.:

$$\sum_{i=2,3,4} \text{Res}_{(p_1+p_i)^2 \rightarrow 0} \left[\text{Diagram with blob} \right] \Big|_{\text{tree}} = - \left[\text{Diagram 1} + \text{Diagram 2} + (t \leftrightarrow u) \right]$$

On-shell EFT amplitudes

3-points

$$\mathcal{M}[1_h^- 2_h^- 3_h^+] = -\frac{1}{M_{Pl}} \frac{\langle 12 \rangle^6}{\langle 13 \rangle^2 \langle 23 \rangle^2}$$

$$\mathcal{M}[1_h^- 2_h^- 3_\phi] = \frac{C_{\phi GB} + iC_{\phi CS}}{M_{Pl}^2} \langle 12 \rangle^4$$

$$\mathcal{M}[1_h^+ 2_h^+ 3_h^-] = -\frac{1}{M_{Pl}} \frac{[12]^6}{[13]^2 [23]^2}$$

$$\mathcal{M}[1_h^+ 2_h^+ 3_\phi] = \frac{C_{\phi GB} - iC_{\phi CS}}{M_{Pl}^2} [12]^4$$

$$\mathcal{M}[1_h^- 2_h^- 3_h^-] = \frac{C_{R^3} + i\widetilde{C}_{R^3}}{M_{Pl}^3} \langle 12 \rangle^2 \langle 13 \rangle^2 \langle 23 \rangle^2$$

$$\mathcal{M}[1_h^- 2_\phi 3_\phi] = -\frac{1}{M_{Pl}} \frac{\langle 12 \rangle^2 \langle 13 \rangle^2}{\langle 23 \rangle^2}$$

$$\mathcal{M}[1_h^+ 2_h^+ 3_h^+] = \frac{C_{R^3} - i\widetilde{C}_{R^3}}{M_{Pl}^3} [12]^2 [13]^2 [23]^2$$

$$\mathcal{M}[1_h^+ 2_\phi 3_\phi] = -\frac{1}{M_{Pl}} \frac{[12]^2 [13]^2}{[23]^2}$$

$$\mathcal{M}[1_\phi 2_\phi 3_\phi] = -C_3$$

Building
blocks

4-points

$$\mathcal{M}[1_h^- 2_h^- 3_h^+ 4_\phi] = -\frac{C_{\phi GB} + iC_{\phi CS}}{M_{Pl}^3} \frac{\langle 12 \rangle^4 [3 | p_1 p_2 | 3]^2}{stu} + \text{contact},$$

$$\mathcal{M}[1_h^- 2_\phi 3_\phi 4_\phi] = \langle 1 | p_2 p_3 | 1 \rangle^2 \left\{ \frac{C_3}{stu} + \frac{C_{\phi GB} + iC_{\phi CS}}{M_{Pl}^3} \left[\frac{1}{s} + \frac{1}{t} + \frac{1}{u} \right] \right\} + \text{contact},$$

$$\mathcal{M}[1_h^- 2_h^- 3_\phi 4_\phi] = \langle 12 \rangle^4 \left\{ -\frac{C_{R^3} + i\widetilde{C}_{R^3}}{M_{Pl}^4} \frac{tu}{s} + \frac{C_{\phi^2 GB} + iC_{\phi^2 CS}}{M_{Pl}^2} \right\} + \dots$$

To be
matched
with the UV

Shift-symmetric limit: $y_p, c_\Psi \neq 0$



$$C_{\phi\text{GB}} = -\frac{\cancel{y_0}}{1440\pi^2 m_\Phi} - \frac{\cancel{7y_s}}{2880\pi^2 m_\Psi} - \frac{\cancel{11y_d - 2y_m}}{240\pi^2 M_*} + \mathcal{O}(M_*^{-1})$$

$$C_{\phi\text{CS}} = -\frac{y_p}{192\pi^2 m_\Psi} - \frac{\cancel{y_a}}{24\pi^2 M_*} + \mathcal{O}(M_*^{-1})$$

$$c_\Psi = \frac{y_p^2 M_*^*}{m_\Psi}, \quad C_{\phi^2\text{GB}} \rightarrow 0$$

Shift-symmetric SGB/DCS interactions

$$C_{\phi^2\text{GB}} = \frac{\cancel{y_0^2} - \cancel{c_\Phi}}{1440\pi^2 m_\Phi^2} + \frac{\cancel{7(y_s^2 - y_p^2)}}{2880\pi^2 m_\Psi^2} + \frac{\cancel{7c_\Psi}}{2880\pi^2 M_* m_\Psi} + \frac{1}{M_*^2 \pi^2} \left\{ \frac{5y_d^2}{96} - \frac{\cancel{y_m^2}}{768} - \frac{\cancel{19y_a^2}}{480} + \frac{11c_V}{240} - \frac{11\hat{c}_V}{1920} \right\} + \mathcal{O}(M_*^{-2})$$

$$C_{\phi^2\text{CS}} = \frac{\cancel{y_s y_p}}{96\pi^2 m_\Psi^2} + \frac{\cancel{\tilde{c}_\Psi}}{192\pi^2 M_* m_\Psi} + \frac{\cancel{2y_d y_a} + \tilde{c}_V}{24M_*^2 \pi^2} + \mathcal{O}(M_*^{-2})$$

$$C_{\phi R^3} = \frac{\cancel{y_0}}{40320\pi^2 m_\Phi^3} - \frac{\cancel{y_s}}{10080\pi^2 m_\Psi^3} + \frac{\cancel{2y_d + y_m}}{13440\pi^2 M_* m_V^2} + \mathcal{O}(M_*^{-3})$$

$$\tilde{C}_{\phi R^3} = \mathcal{O}(M_*^{-2})$$

$$C_{R^3} = -\frac{1}{40320\pi^2 m_\Phi^2} + \frac{1}{20160\pi^2 m_\Psi^2} - \frac{1}{13440\pi^2 m_V^2} + \mathcal{O}(M_*^{-2})$$

$$\tilde{C}_{R^3} = \mathcal{O}(M_*^{-2})$$

Wilson coeffs. for R^3 interaction
Crosschecked with

J.Phys.A 54 (2021) 34, 344002 [Bern, Kosmopoulos, Zhiboedov]
Eur.Phys.J.C 85 (2025) 2, 154 [Alviani, Falkowski]

Shift-symmetry
breaking
interactions

Comments

1. All predictive theories that generate $C_{\phi\text{GB}}$ also generate $C_{\phi^2\text{GB}}$ with a pattern consistent with exponentiation:

$$\sim e^{C_{\text{GB}}\phi} \mathcal{R}_{\text{GB}}$$

Arises from string theory models, with ϕ the dilaton
Does exponentiation indeed (always) arise?

2. One can notice that some of our UV interactions are shift-symmetric, i.e. lead to vanishing amplitudes for $p_\phi \rightarrow 0$, e.g.

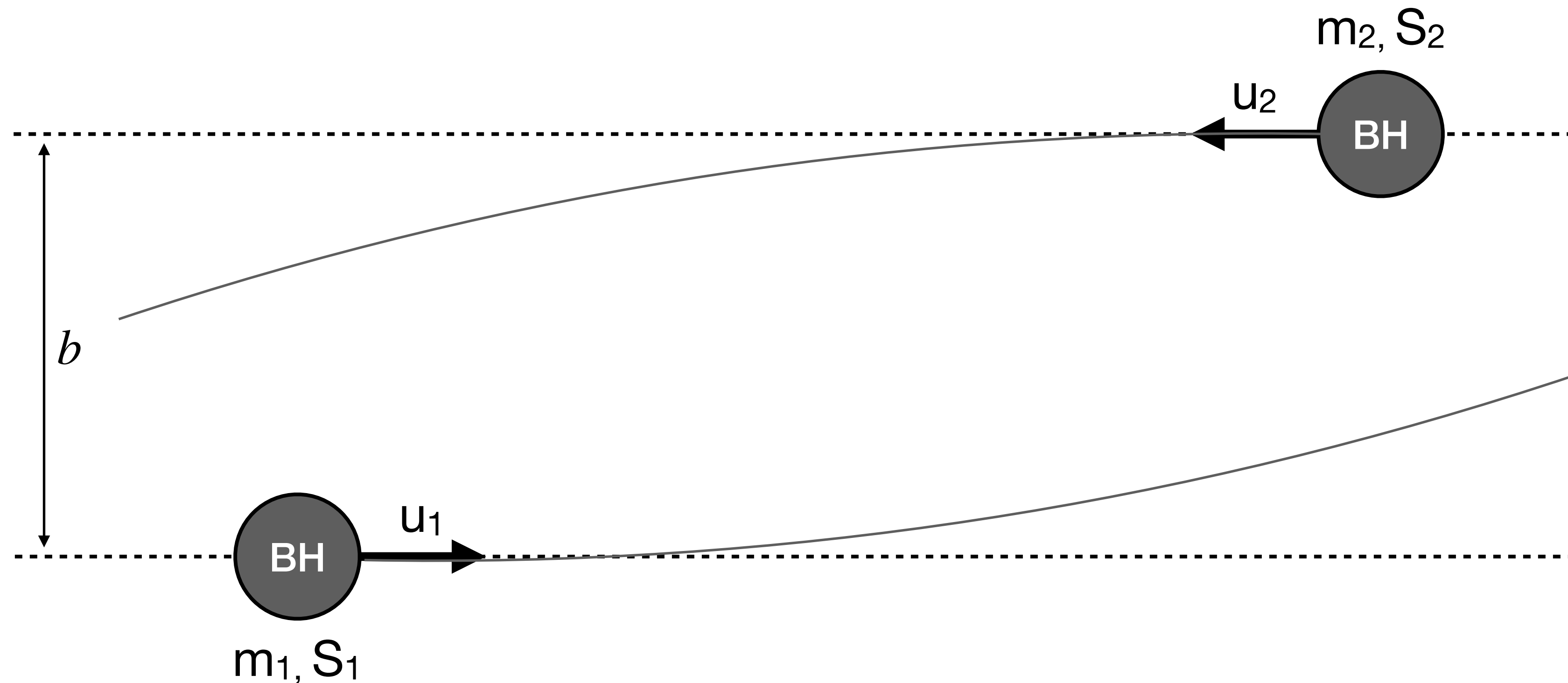
$$\mathcal{M}[1_\phi 2_\Psi 3_{\bar{\Psi}}] = -iy_p [\langle \mathbf{23} \rangle - [\mathbf{23}]] \sim p_\phi$$

$$\mathcal{M}[1_\phi 2_V 3_{\bar{V}}] = -\frac{iy_a}{M_*} [\langle \mathbf{23} \rangle^2 - [\mathbf{23}]^2] \sim p_\phi$$

Why aren't these interactions alone preserve the shift-symmetry in the IR?



2. Scattering Amplitudes and Observables



Weak field expansion:

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2}{M_{Pl}} h_{\mu\nu}$$

PM expansion:

$$R_s \ll b$$

Spin expansion:

$$\frac{S}{m} \ll b$$

- **Focus: Classical scattering problem in GR**

→ Can we describe this problem using the Scattering Amplitudes used in QFT?

Yes! → Use of the **KMOC formalism**



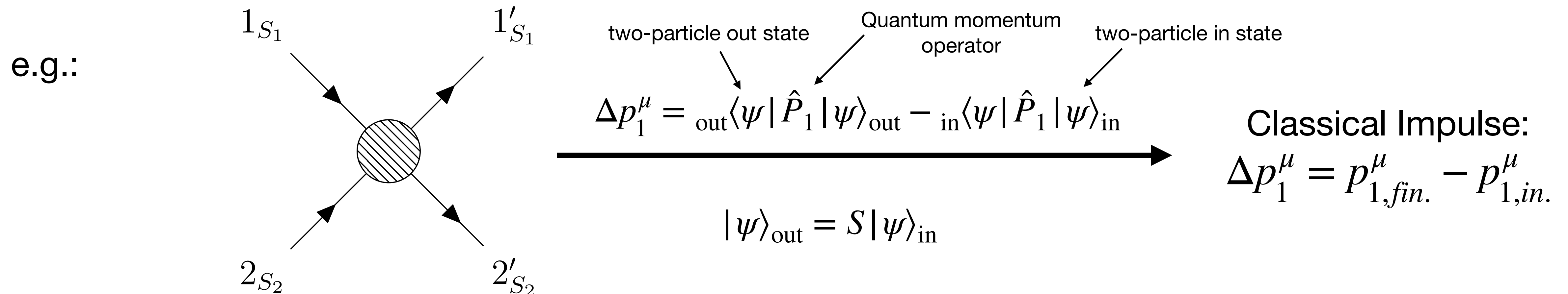
KMOC (Kosower Maybee O'Connell) formalism

JHEP 02 (2019) 137 [Kosower, Maybee, O'Connell]

Phys.Rev.D 106 (2022) 5, 0567007 [Cristofoli, Gonzo, Kosower, O'Connell]

Idea: Relate **Scattering Amplitudes** directly to **classical observables**

→ Extract the classical piece of the amplitude through an “ $\hbar$ ” counting prescription



At leading order:
$$\Delta p_1^{\mu, LO} = \frac{i}{4} \left\langle \left\langle \hbar^2 \int \hat{d}^4 q \hat{\delta}(q \cdot p_1) \hat{\delta}(q \cdot p_2) e^{-ib \cdot q} q^\mu \mathcal{M}^{LO}(p_1, p_2 \rightarrow p_1 + \hbar q, p_2 - \hbar q) \right\rangle \right\rangle$$

Waveforms from amplitudes

Similar approach can be applied to radiation observables

Examples:

Massless scalar theory

$$R_\phi \equiv {}_{\text{out}}\langle\psi|\phi(x)|\psi\rangle_{\text{out}}$$

Electromagnetism

$$R_\mu \equiv {}_{\text{out}}\langle\psi|A_\mu(x)|\psi\rangle_{\text{out}}$$

or gauge invariant

$$R_{\mu\nu} \equiv {}_{\text{out}}\langle\psi|F_{\mu\nu}(x)|\psi\rangle_{\text{out}}$$

$$R_A^\pm \equiv {}_{\text{out}}\langle\psi|A^\mu(x)|\psi\rangle_{\text{out}}\epsilon_\mu^\pm$$

GR

$$R_{\mu\nu} \equiv {}_{\text{out}}\langle\psi|h_{\mu\nu}(x)|\psi\rangle_{\text{out}}$$

or gauge invariant

$$R_{\mu\nu\alpha\beta} \equiv {}_{\text{out}}\langle\psi|R_{\mu\nu\alpha\beta}(x)|\psi\rangle_{\text{out}}$$

$$R_h \equiv {}_{\text{out}}\langle\psi|h^{\mu\nu}(x)|\psi\rangle_{\text{out}}\epsilon_{\mu\nu}^-$$

Waveforms from amplitudes

Given radiation observable R_X one defines waveform W_X as

$$R_X(x) = \frac{W_X(t)}{|\mathbf{x}|} \quad |\mathbf{x}| \rightarrow \infty \quad t \equiv x^0 - |\mathbf{x}|$$

retarded time

Furthermore one defines spectral waveform f_X as Fourier transform:

$$f_X(\omega) = \int dt \, e^{i\omega t} W_X(t)$$

KMOC formalism relates spectral waveform at leading order to integral of 5-point amplitude.

At leading order:

$$f_X(\omega) = \frac{1}{64\pi^3 m_1 m_2} \int d\mu \mathcal{M}^{\text{cl}}[p_1 + w_1, p_2 + w_2 \rightarrow p_1, p_2, k] \Big|_{k=\omega n}$$

Integration measure

$d\mu \equiv \delta^4(w_1 + w_2 - k) \Pi_{i=1,2} [e^{ib_i w_i} d^4 w_i \delta(u_i w_i)]$

Matter representing
classical bodies

Quantum of particle X

Waveforms at leading order:

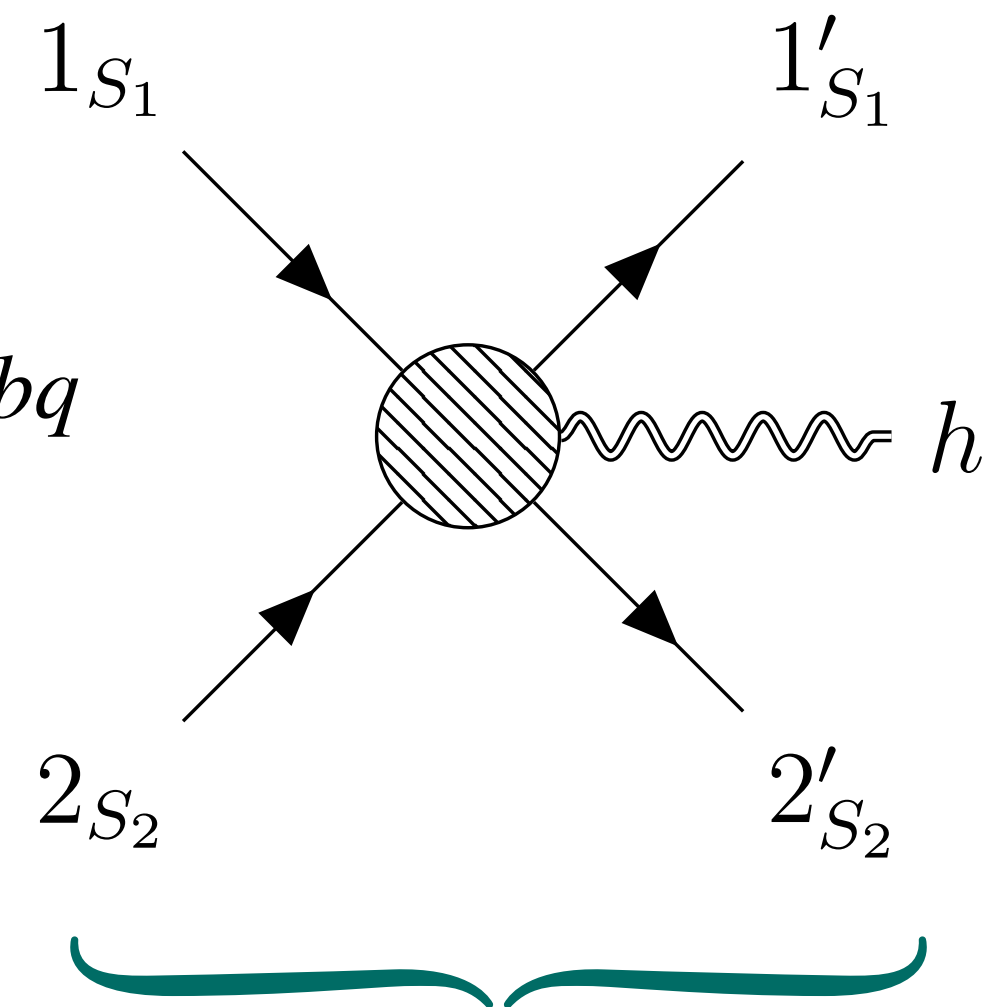
JHEP 02 (2019) 137 [Kosower, Maybee, O'Connell]
 Phys.Rev.D 106 (2022) 5, 0567007 [Cristofoli, Gonzo, Kosower, O'Connell]

$$R_{GR}(t) \sim \frac{h_{GR}(t)}{|x|}, \quad |x| \rightarrow \infty, \quad \text{retarded time } t \equiv x^0 - |x|, \quad R_{GR} \equiv {}_{\text{out}}\langle \psi | h^{\mu\nu}(x) | \psi \rangle_{\text{out}} \epsilon_{\mu\nu}^-$$

strain

on-shell measure

$$h_{GR}(t) \equiv h_+ \pm i h_\times \sim \int d\omega e^{-i\omega t} \int d\Phi(q) e^{-ibq}$$



$$\mathcal{M}_5^{\text{cl}}(q, k) |_{k^\mu = \omega n^\mu}$$

But, why?



- Computations organized in a **perturbative expansion** using a **simple algorithm**.
- **Analytic results**, in places where either PN approximations or NR was used before.
- Can exploit many **modern techniques used in particle physics** to simplify the problem.
- Can straightforwardly **extend to beyond GR predictions**

Resolvability (no-hair compact objects):

Scalar radiation for no-hair objects is resolvable if: $\alpha_g \left(\frac{R_s}{b} \right)^2 \frac{1}{\Lambda^4 b^4} \gg 1$,

$$\alpha_g \sim \frac{m^2}{M_{Pl}^2} = m R_s$$

For example, for $\Lambda \simeq 2 \times 10^{-18} GeV$ and $m \simeq M_\odot \simeq 10^{57} \sim GeV$ (corresponding to $R_s \simeq 1.5 \times 10^{19} GeV^{-1}$) we have $b_{\max} \sim 10^{11} R_s$

Linear in spin order effects for the same observable: $\alpha_g \left(\frac{R_s}{b} \right)^2 \frac{1}{\Lambda^4 b^4} \left(\frac{S}{bm} \right) \gg 1$

For our benchmark point, $m \sim M_\odot, \Lambda \sim 0.1 km^{-1}$, this implies $S \gtrsim 10^6 (b/R_s)^7$, with $S_{\max} \sim 10^{76}$ for Kerr black holes in GR.

Details on the matching for static SGB BHs

Phys.Rev.D 100 (2019) 10, 104061
[Julié, Berti]

$$\begin{aligned}
 ds^2 &= -A(r)dt^2 + \frac{dr^2}{A(r)} + B(r)r^2(d\theta^2 + \sin^2\theta d\phi^2) & \phi(r) &= \phi_\infty + \sum_i \epsilon^i \phi_i(r, m) \\
 A(r) &= 1 - \frac{2m}{r} + \sum_i \epsilon^i A_i(r, m) & \epsilon &\sim \alpha \frac{M_{Pl}^4}{\Lambda^2 m^2} \\
 B(r) &= 1 + \sum_i \epsilon^i B_i(r, m)
 \end{aligned}$$

1. Build perturbative BH solution for both metric and scalar field

$$\begin{aligned}
 S &= S_{bulk} + S_m, \quad S_m = \int m_A(\phi) ds_A, \quad ds_A = \sqrt{-g_{\mu\nu} dx_A^\mu dx_B^\nu} \\
 \tilde{g}_{\mu\nu} &= \eta_{\mu\nu} + \delta_{\mu\nu} \left(\frac{2m_A(\phi_\infty)}{\tilde{r}} \right) + \mathcal{O}\left(\frac{1}{\tilde{r}^2}\right), \quad \phi = \phi_\infty - \frac{1}{\tilde{r}} \frac{dm_A}{d\phi}(\phi_\infty) + \mathcal{O}\left(\frac{1}{\tilde{r}^2}\right)
 \end{aligned}$$

2. Build skeletonized (wordline) action

$$m_A(\phi_\infty) = m, \quad m'_A(\phi_\infty) = -c_A, \quad \dots$$

the monopole scalar charge

3. Match the two descriptions:
 $c_A = c_A(\epsilon)$

A modified BHPT framework to match ala
JHEP 05 (2023) 211 [Bautista, Guevara, Kavangh, Vines] ?

Inclusion
of spin:

Phys.Rev.D 110 (2024) 12, 124016 [Almeida, Zhou]